UNIVERSITY OF AUCKLAND
Waipapa Taumata Rau
NEW ZEALAND

SCIENCE

COMPSCI 714
Lectutorial 5: Learning with images - CNNs & ViT

How to implement and use pretrained CNNs + Principles and use of the Vision Transformer

Instructor: Thomas Lacombe

Learning outcomes

- **Apply and interpret 1D convolution** for sequential data (including dilated/causal convolutions), and **explain** its role in models such as WaveNet for extending convolutional methods beyond images.

- **Implement, train, and evaluate** CNN models in PyTorch for image-based learning tasks.

- **Load and apply** pre-trained CNN and ViT models to image-based learning tasks.

Extra resources

If you want to go further:

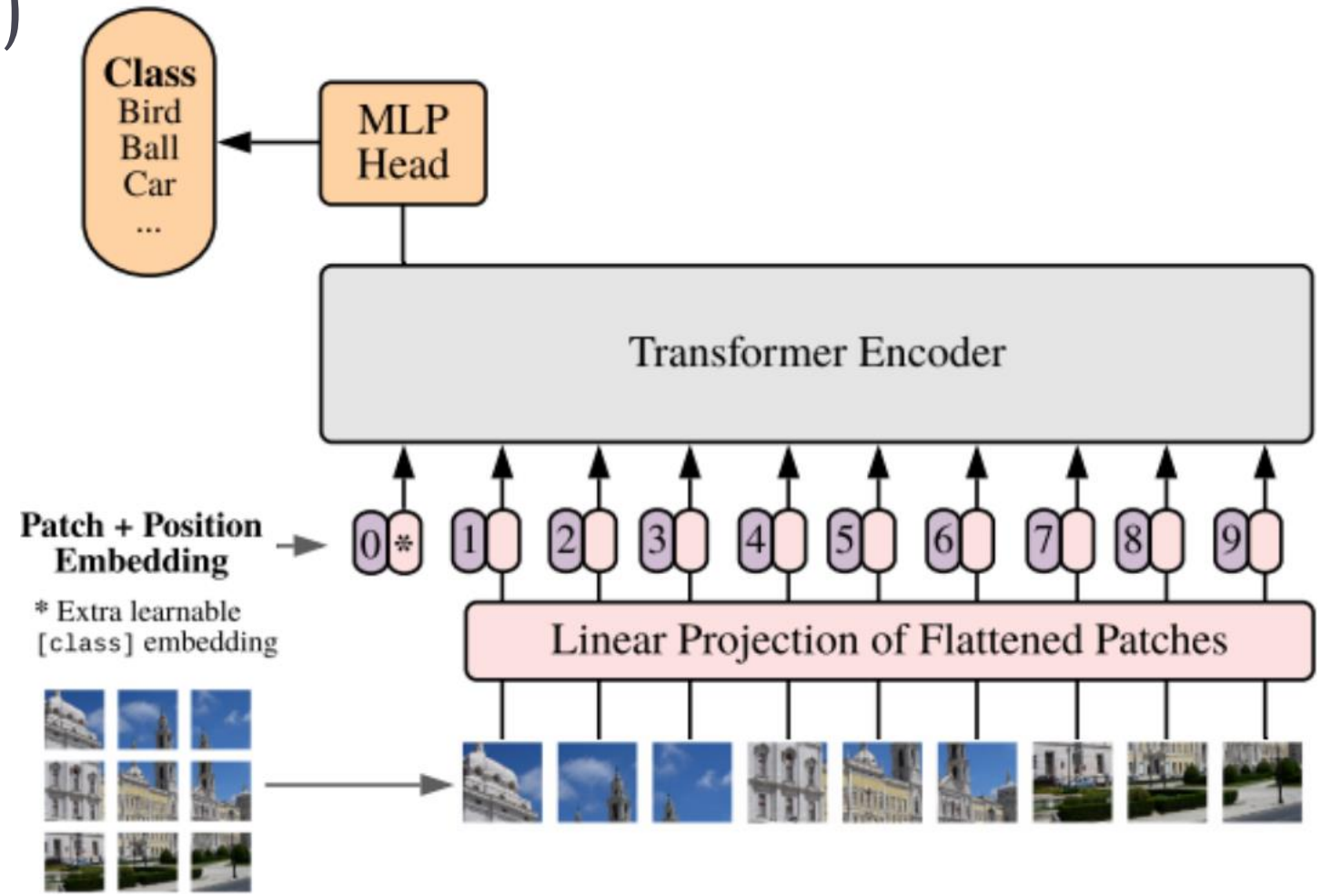
- *Hands-on Machine Learning with Scikit-Learn and PyTorch*, Géron A.
  - Chapter 13 and 16.
- *Deep Learning*, Goodfellow I. et al.
  - Part II Deep Networks, chap. 9.

Acknowledgement:
The code in the notebook is adapted from the GitHub repository: ageron/handson-mlp.

Coding time 1: Convolutional Neural Networks (CNNs)

Vision Transformer (ViT)

- Proposed by Google Brain team in 2020.
- Chops the image into 16x16 squares/patches (flattened).
- Treat the sequence of squares as if it was a sequence of word embeddings.
- Position embedding.
- Use the transformer encoder to generate a representation that is fed to a MLP head for classification.



[Vision Transformer, 2020]
https://blog.research.google/2020/12/transformers-for-image-recognition-at.html

Vision Transformer (ViT)

- The ViT was able to beat SoTA CNN models on the ImageNet image classification task (millions of images from 1000 different classes).

- However, it needed more training data (over 300 million additional images), which makes sense as it has less *inductive biases* than a CNN when working with images.
  - *Inductive biases* refer to implicit assumptions made by a model due to its architecture (e.g., CNN implicitly assumes that patterns learned at one location will likely be useful in another locations, because of the use of convolutional layers).

- Since 2020, a lot of developments happened with ViT, see "Hands on ML", chap. 16, and survey "A Survey on Visual Transformer" (2023).

Vision Transformer (ViT) vs CNN

- **Intuition**
  - CNN: “Scan the image with a small detector”
  - ViT: “Compare every patch with every other patch”

- CNNs rely on strong built-in assumptions and work well with limited data.
- ViTs are more flexible and powerful at scale but require more data and compute.

- **Use CNNs when:**
  - Dataset is small to medium
  - You want efficient training/inference
  - Task relies on local patterns (e.g., medical imaging, defect detection)

- **Use ViTs when:**
  - You have large-scale data or pretrained models
  - Task benefits from global context (e.g., complex scenes)
  - You want state-of-the-art performance at scale
  - (Can also use distillation to train a ViT more efficiently)

Touvron, Hugo, et al. "Training data-efficient image transformers & distillation through attention." International conference on machine learning. PMLR, 2021.

Coding time 2: ViT VS ResNet using Hugging Face

Extra material slides (not evaluated)

Convolutions for Sequential Data

- 1 D Convolution
- WaveNet
- (See extra notebook for implementation with TensorFlow/Keras)

1D Convolution

Output	4											

Filter (1x1)
2

Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	4	6										

Filter (1x1)	2

Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	4	6	14									
Filter (1x1)			2									
Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	4	6	14	2	14	8	18	20	4	10	16	4

Filter (1x1)

2

Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

Output	**35**									

Filter (1x3)	3	5	2

Input	**2**	**3**	**7**	**1**	**7**	**4**	**9**	**10**	**2**	**5**	**8**	**2**

1D Convolution

Output	**35**	**46**									

Filter (1x3)	3	5	2

Input	**2**	**3**	**7**	**1**	**7**	**4**	**9**	**10**	**2**	**5**	**8**	**2**

1D Convolution

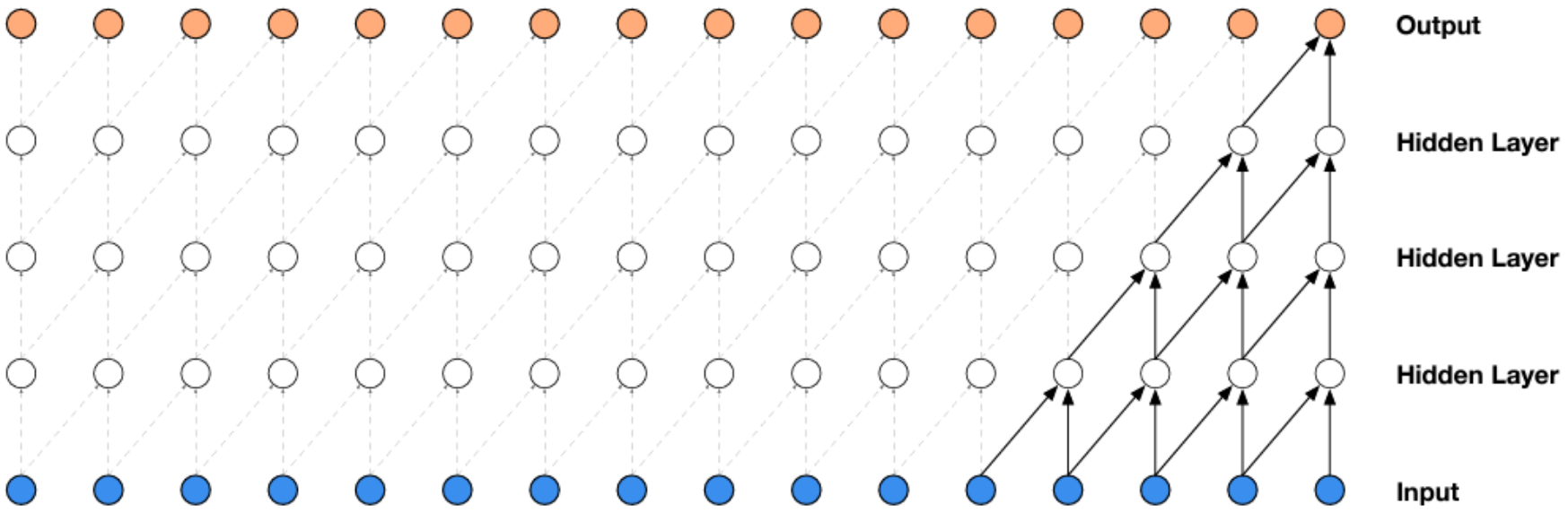
Output		35	46	40	47	59	67	81	50	47	59	
Filter (1x3)										3	5	2
Input	2	3	7	1	7	4	9	10	2	5	8	2

1D Convolution

- When is it useful?
  - **Train 1D convolution layer on sequential data**: each kernel in the layer will learn a single short sequential pattern (no longer than the kernel size)
  - Can be used to learn features on the sequence, but also to downsample the sequence (if valid padding or stride > 1).
  - Can be combined with RNN layers (see extra material notebook).

WaveNet

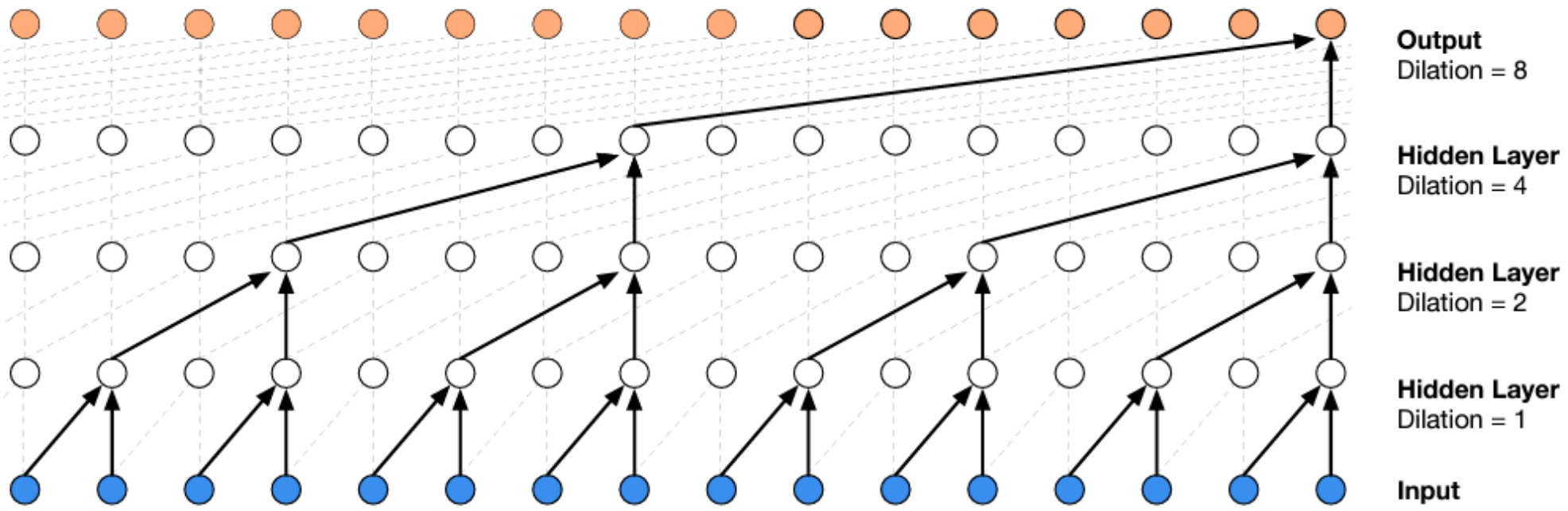
- Network proposed in 2016 by DeepMind/Google for audio waveform generation
  - Stack 1D causal convolution layers
  - Causal convolution means that a neuron is connected to current and previous time steps (not future).



[WaveNet, 2016]
https://deepmind.google/discover/blog/wavenet-a-generative-model-for-raw-audio/

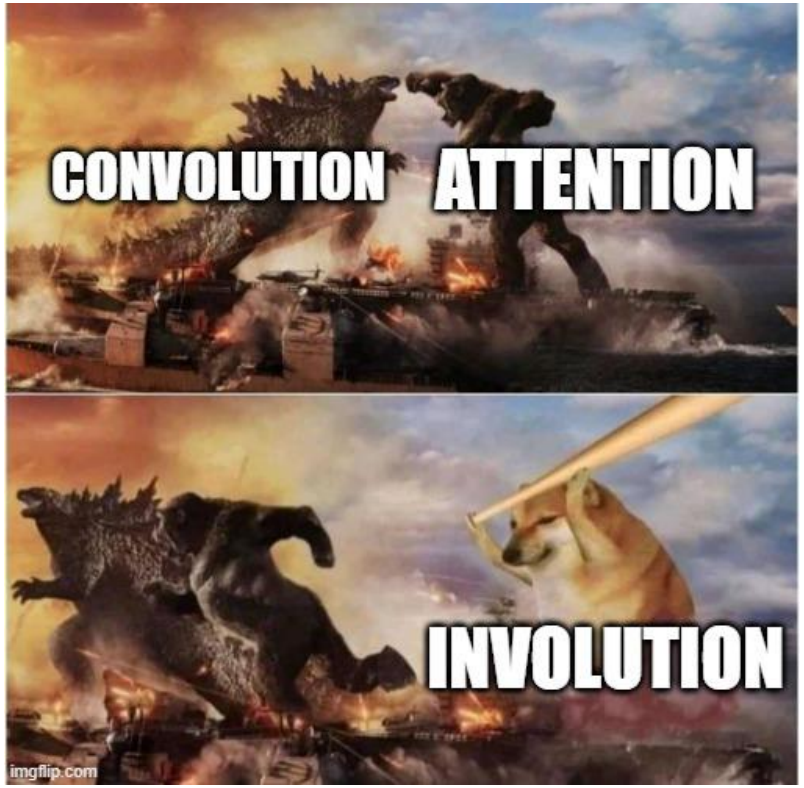
Wavenet

- Causal dilated convolution
  - First hidden layer sees 2 input time steps, next one sees 4, the next one sees 8, etc.
  - Lower layers learn short term patterns while deeper layers learn long-term patterns.



[Wavenet, 2016]
https://deepmind.google/discover/blog/wavenet-a-generative-model-for-raw-audio/

Conclusion meme



https://keras.io/examples/vision/involution/